Gerardo

Summary:

Key Experience1: 22+ years of experience in programming and 15+ years of experience building enterprise software applications in different industries like Medical, Financial, Government, Consultancy, and Retail.

Key Experience2: Expertise in Software Development Life Cycle: planning, defining requirements, design & prototyping, development, testing, deployment, operation & maintenance using Agile (Scrum), Waterfall, and Test-driven development (TDD).

Key Experience3: 3+ years of experience with Microservices Stack (Configuration management, Service discovery, Fault tolerance, Centralized logging, Centralized metrics, Distributed tracing, Scheduling, Deployment, Auto Scaling).

Key Experience4: 3+ years of experience with Cloud Technologies like Google Cloud Platform, Azure, Kubernetes, Docker, and API Gateways (Apigee, Layer7).

Professional Summary:

Proficient with integration tools like IBM Integration Bus (IIB/ACE) and Rational Software Architect (RSA) for enterprise system design.

Skilled in Java 11 and beyond, Java Spring framework, REST API development, AWS cloud services, Angular JavaScript for front-end development, and SQL for relational databases.

- Experience with Reactive Programming and Functional Programming.
- Extensive expertise with:

  Architecture & Integration: Rational Software Architect (RSA), IBM Integration Bus (IIB/ACE), Microservices, Design Patterns (DAO, DTO, Singleton, Abstract Factory, Business Delegate, CQRS, Saga, Circuit Breaker).

  Core Skills: Java 11 or beyond, Spring Framework (Spring Boot, Spring MVC, Spring Data, Spring Security), REST API design and implementation, AWS (EC2, S3, RDS, Lambda), Angular JavaScript, SQL (Oracle, PostgreSQL, MySQL).
- Java 11 (Type Inference); Java 8 (Streams and Lambda expressions); Java 6 (Collections); Java 5 (Generics); Java
- 4(XML support).
- Responsible for Development of Web page using HTML5, JavaScript, jQuery,React.jsImmutable JS,CSS, SASS/SCSSincluding Ajax controls and JSON.
- The J2EE platform like Servlets, Java Server Pages (JSP), Java Database Connectivity (JDBC), Java Persistence API (JPA), Enterprise JavaBean (EJB), Java Message Service API (JMS), Security, Java Transaction API (JTA), Java API for XML Web Services (JAX-WS), Java API for RESTful Web Services (JAX-RS).
- Java Frameworks like Spring (Reactive, Boot, MVC, Data, Security, Cloud, Apache Kafka, Web Services), Hibernate, Java Server Faces (JSF), Struts.
- Web/Application Servers like Tomcat (v8, v9), Oracle WebLogic, IBM WebSphere, Red Hat WildFly, and Oracle GlassFish for deploying and configuring web applications.
- Familiar with REST API and GRAPHQL.
- Web Design like Angular, HTML, HTML Document Object Model (DOM), XHTML, Extensible Markup Language (XML), Extensible Stylesheet Language (XSLT), Cascading Style Sheets (CSS),

Gerardo

JavaScript, Asynchronous JavaScript and XML (AJAX), jQuery, JavaScript Object Notation (JSON).
- Relational Database Management System (RDBMS) like Oracle 12g, SQL Server, MySQL, PostgreSQL using queries, PL-SQL, stored procedures, functions, and views.
- NoSQL databases like MongoDB, Google Firestore, and Azure Cosmos.
- Messaging tools like Apache Kafka.
- Enterprise Service Bus (ESB) like TIBCO.
- Testing frameworks like JUnit, Mockito, and REST-assured.
- Development tools like Eclipse, IntelliJ, NetBeans, SQL Developer, Postman, SOAP-UI, NoSQLBooster, Robomongo, and TOAD.
- Workflow and Decision Automation Platform tools like Camunda.
- Project management tools like Maven.
- Text editors like Sublime Text and Notepad++.
- Proficient in the integration of tooling for CI/CD like Nexus Repository Manager, Sonar, Looper, and Jenkins
- Work with version control systems like SVN, Git, and GitHub.
- Project Tracking Software like Jira
- Strong Core Java with a strong understanding of principles of Object-Oriented paradigm: Encapsulation, Abstraction, Inheritance, Polymorphism; Multi-threading, Exception handling, Garbage Collector, Design patterns (Data Access Object, Data Transfer Objects, Abstract Factory, Service Data Objects, Business Object, Singleton, Interfaces, Business Delegate), Unified Modeling Language (UML).
- Learn Python to work with analysis of Big Data.
- Learn Scala to improve Functional Programming concepts.
- Job experience like Sysadmin on UNIX, and Linux platforms, developed shell scripts, and configured CRON jobs.

Education:
- National Polytechnic Institute Mexico City 2001- 2006
- Bachelors in Computer Systems Engineering

Professional Experience:

Family Savings and Loan Cooperative, Remote **Jan 2025 – current**

- Developed and deployed microservices using Java 11 and Spring Boot with REST APIs on AWS cloud.

- Designed database schema and implemented SQL queries for data persistence.

- Collaborated with UI teams leveraging Angular JavaScript for seamless front-end integration.

Role: Java Solutions Architect

Description: Savings and Loan Cooperative is moving its financial system from a monolithic legacy to cloud architecture. As a solution architect, I am responsible for designing and implementing the business logic oriented to a native cloud application, selecting appropriate technologies that are scalable, secure, and maintainable over time, and aligned with the organization's strategic objectives.

Responsibilities as Architect:
- Work closely with stakeholders to understand the requirements, constraints, and goals.
- Translate functional and non-functional requirements into a technical solution.
- Evaluate technologies to determine the best fit based on business requirements.
- Design the solution, deliver technical specifications and architecture diagrams, and
- define software, hardware, and network components.

Gerardo

- Worked in React JS for creating interactive UI& using One-way data flow, Virtual DOM, JSX, React concepts
- Implemented React router for building the single page application.
- Used props, states, keys, refs in React.js to build the UI components.
- partials through RESTFUL services from the server side and handling view logic on the client side.

Environment: AWS, Google Cloud Platform, Architectural Patterns.

Equifax, Atlanta, GA -Active Alliance / Mindtree **Oct 2022 – Apr 2025**

- Worked extensively with Java 11 and Spring Boot to design and implement REST APIs.
- Integrated Angular JavaScript components with backend APIs for full-stack delivery.
- Utilized AWS services for hosting and deployment of microservices.
- Designed SQL queries for relational data persistence and reporting.

Role: Technical Leader / Full Stack Developer

Description: Equifax consolidates information from multiple partners to create financial products for its clients. Due to its expansion, I aim to develop the necessary plugins to clean, import, and integrate the partners' data into Equifax's infrastructure, moving mainframe-based business logic to the Cloud environment, adhering to its standards and architecture. Besides, it has an area called Workforce Solutions to develop tools like "I-9 HQ" designed to help Human Resources teams and new hires save their I-9 Form information and stay in contact during the hiring process.

Responsibilities as Technical Leader:
Understand Equifax´s business logic and the technological core to make it comprehensible to the developers; this includes understanding Equifax's tools (ETL process) developed under the Google Cloud Platform.
Be responsible for leading a group of developers and testers in creating the Equifax configuration files and Java files necessary to load source information into GCP. The technology used here is Apache Avro and internal Equifax APIs developed under the Google Cloud Platform.
Document and explain all internal processes to make the logic used in Equifax understandable to programmers.
Improve the skills of developers and QA testers.

Responsibilities as Full Stack Developer:

- Understand I-9 Web Portal business logic developed under Angular, Equifax APIs, and Google Cloud Platform.
- Create and modify Angular components to respond to new business requirements in the web portal, such as uploading files, making HTTP calls, and updating UI.
- Developed user interface by using the React JS, Flux/Redux for SPA development.
- Involved in using ReactJS components, Forms, Events, Keys, Router, Animations and Flux concept.
- Create Angular test scenarios related to Angular components.
- Use Jenkins CI/CD pipelines to deploy Angular code into environments.

Environment: Java 11, Google Cloud Platform (Kubernetes Engine, Cloud Storage [Buckets], Big Query, Pub/Sub, Dataflow, Airflow, Data prep, Fire store Database, BigTable Database), Apache Avro, Angular, Internal Equifax APIs.

Gerardo

Adobe, Houston TX -Active Alliance / Nextgentech **Jan 2022 – Sep 2022**

- Developed services with Java 18 (beyond Java 11) and Spring Boot microservices with REST APIs.
- Implemented SQL-based queries to handle structured datasets and optimize performance.
- Ensured microservices deployment compatibility with AWS environments.

Role: Software Developer Java 3

Description: Adobe maintains hundreds of millions of records related to User Experience from its clients. Due to storage being generally an expensive resource, my objective was to create a first-approach statistics of its Big Data to save storage and see correlations between data.

Responsibilities:

- Developing Rest services with Java 18 and Spring Boot as microservices and improving internal Java projects dedicated to testing the User Experience pipelines.
- To Learn the meaning of the data.
- To create a first-approach solution to visualize the correlations between the data graphically using Python, Pandas
- (a Python library for working with data sets), Matplotlib (a low-level graph plotting Python library), Java, and Gradle.

Environment: Java 18, Python 3, Gradle.

EY (Ernst & Young), Houston TX- Active Alliance / Pyramid **Jul 2021 – Oct 2021**

- Built and tested microservices using Java 11 and Spring Reactive with REST APIs.
- Applied SQL-based validation for backend processing and reporting.
- Collaborated with teams to ensure AWS integration for reactive workloads.

Role: Reactive Microservices developer

Description: EY, one of the Big Four accounting firms, is developing reactive event-driven microservices to perform internal tasks to create executive reports for their clients. As a reactive microservices developer using Spring Reactive technology, my mission was to validate the performance of the microservices through load tests. Update the business decision rules implemented in Camunda. Create unit test cases to validate business logic behavior. The tech stack includes Azure (Portal and DevOps), Camunda, Apache Kafka, Azure Cassandra, and Azure Databricks.

Responsibilities:

- Create test cases to validate reactive programming and functional programming methods.
- Modify Camunda's Business Process Model and Notation (BPMN) and Decision Model and Notation (DMN) files.
- Give support load testing phases, including creating internal statistics and error analysis.
- Update microservices building knowledge using Spring Reactive Flux and Mono objects.
- The first approach to Python as a language preferred to create Azure Databricks notebooks.
- Use Azure DevOps as a code repository, CI/CD, and planning tool.
- Use Azure Portal as PaaS

Gerardo

Environment: IntelliJ, Java 11, Spring Reactive, Spring Boot, Kafka, Azure Cassandra, Postman, Git, Maven, Junit.

Client: Walmart USA, Mexico City, Softtek **Mar 2020 – Jun 2021**

- Implemented integration workflows with IBM Integration Bus (IIB/ACE) for secure data flows.
- Applied RSA to model distributed architecture and ensure alignment with enterprise standards.
- Designed microservices architecture using Java Spring Boot and REST API patterns.
- Migrated legacy systems into AWS cloud-based infrastructure.
- Developed SQL queries to validate migrated transactional data.

Architect / Technical Lead / Microservices developer

Description: Walmart USA, the biggest retail company globally, is migrating its Mainframe business core to the Cloud in an effort called MF2C. As the team leader, my team and I were assigned to create the architecture and develop a complete cloud-based solution to migrate the mainframe-based flow using new and legacy microservices, Kafka, NoSQL, and PaaS as the technical stack.

- Implemented integration workflows with IBM Integration Bus (IIB/ACE) for secure data flows.
- Applied RSA to model distributed architecture and ensure alignment with enterprise standards.

Responsibilities:

- Understand the core business logic at high (Walmart), medium (Accounts Payable), and low (Store Receiving's) levels.
- Analyze the issues related to using two different programming paradigms and their contrasting conceptions to organize the data: procedural/relational vs. object-oriented/non-relational.
- Led a team of 4 developers, responsible for task allocation, mentoring, best code practices, issue resolutions, pair programming, and architecture design.
- Generate the Architecture using microservices, queues, and transaction logic to represent the core logic implemented on the mainframe side.
- Migrate mainframe logic through reading mainframe code to create a high-level specification to be implemented in Java.
- Design the Representational State Transfer (REST) modules: data collection, data enrichment, store-receiving's core, confirmations, and reports.
- Validate the data migration from a Relational Database to a NoSQL database performed by another team.
- Developing Rest services with Java 8 and Spring Boot as microservices, these web services were deployed in Walmart Cloud Native Platform (WCNP) infrastructure.
- Develop the messages and payloads to interchange data between the services.
- Implement Kafka consumers and producers to use the asynchronous messaging and make a fault-tolerant solution.
- Use Git as a code repository.
- Use Maven as a management tool.
- Use Azure Cosmos as a NoSQL database.
- Use Looper as a tool for CI/CD
- Use JUnit, REST-assured as the testing framework.

Gerardo

Environment: IntelliJ, Java 8, Spring Boot, Kafka, DB2, Azure Cosmos, Postman, NoSQL Booster, Robomongo, Git, Maven, Jenkins, JUnit, REST-assured.

Softtek Microservices Academy, Mexico City

- Trained developers on Java 11+, Spring Boot, REST APIs, and SQL best practices.
- Guided students on deploying applications into AWS environments with CI/CD pipelines.

Company: Softtek **Oct 2019 – Feb 2020**

Role: Lead Instructor of Microservices Academy

Description: Softtek is convinced that developing the technical skills of new talent is the key to achieving the organization's objectives. Therefore, my objective as Microservices Lead Instructor was, in 4 months, 80% of participants must have the necessary knowledge to develop business applications oriented to the Microservices Architecture.

Responsibilities:

Led a team of 2 instructors to create the material and labs to teach and develop the technical knowledge of 20 beginner programmers in MS Architecture.

Topics:

- Git as code repository
- Maven as a project management tool
- Domain-Driven Design
- Unified Modeling Language (UML)
- Java basics (abstraction, encapsulation, inheritance, polymorphism, threads)
- Java 8 (Lambdas, Streams)
- Hypertext Markup Language (HTML)
- Servlets & Java Server Pages (JSP)
- Java Database Connectivity (JDBC)
- Persistence and Object-Relational Mapping (ORM)
- Object-Oriented Patterns Design (Creational: Abstract Factory, Singleton. Structural: Composite, Decorator, Facade)
- Spring MVC
- Junit as a Unit Testing tool
- Spring Security
- Microservices fundamentals
- Configuration management like Netflix Archaius,
- Service discovery like Eureka,
- Fault tolerance (Circuit breaker) like Hystrix,
- Centralized logging like Elasticsearch, Logstash, Kibana,
- Centralized metrics,
- Distributed tracing like Splunk,
- Scheduling & deployment like Jenkins pipelines,
- Auto-scaling like Docker and Kubernetes.
- Microservices Design Patterns
- Aggregator,
- API Gateway,

Gerardo

- Asynchronous Messaging,
- Command and Query Responsibility Segregation (CQRS),
- Saga,
- Circuit Breaker.
- Spring Boot
- REST services
- Spring Cloud
- Swagger
- Postman
- Message Brokers
- ApacheMQ
- OAuth 2.0
- API Managers (apigee)
- Repository Managers (Nexus)
- Containers & Images (Docker)
- Continuous Integration (Testing, Build Images, Push Artifacts)
- Continuous Deployment (Jenkins, Kubernetes)
- Developed the exercises to evaluate the participants.

Environment: Java 8, JSP, servlets, Tomcat, Spring Boot, Spring Cloud, Swagger, Postman, ApacheMQ, Git, OAuth 2.0, Maven, UML, Bitbucket, Hibernate, Apigee, Nexus, Heroku Cloud Application Platform, Jenkins, Docker, IntelliJ.

Brunswick, Frisco TX- Softtek **Mar 2019 – Sep 2019**

- Migrated monolith to microservices using Java Spring Boot and REST APIs.
- Developed SQL queries for Oracle 12g persistence.
- Explored AWS integration during migration.

Role: Microservices Developer

Description: Brunswick is trying to move its Monolith Architecture to a new Microservices Architecture. My objective was to migrate a Java backend project to a microservices project using the best practices to demonstrate the benefits of Microservices Architecture.

Responsibilities:

- Responsible for analyzing the legacy Java source code.
- Migrate the legacy source code to a standard convention over configuration (Maven) of Microservices Architecture.
- Create the java project to include Spring Boot, Spring Data, JPA, Azure Insights connections, and the Oracle database.
- Developed and executed Unit Test plans using JUnit and REST-assured, ensuring that results were documented and reviewed with the client.
- Create a blueprint document to explain the use of the Java tech stack.

Environment: Java 1.6-1.8, J2EE, Spring Boot, Spring Data, JPA, Hibernate 4, Maven, JUnit, REST-assured, IntelliJ, Oracle 12g, Scrum.

Gerardo

MoneyGram, Frisco TX- Softtek **Oct 2018 – Feb 2019**

- Applied RSA for integration modeling and architecture documentation.
- Collaborated on migration from Tibco ESB to Java microservices with IIB/ACE exposure.
- Migrated legacy code into Spring Boot microservices with REST APIs.
- Implemented SQL persistence for distributed services.
- Introduced AWS migration strategies for improved scalability.

Role: Technical Leader

Description: MoneyGram has a big part of its infrastructure in split modules of different technologies and vendors, maintaining communication through an Enterprise Service Bus. The objective was to migrate distributed backend services to microservices. Update deployment process to CD/CI approach and improve user experience by developing new functionality using its tech stack.

- Applied RSA for integration modeling and architecture documentation.
- Collaborated on migration from Tibco ESB to Java microservices with IIB/ACE exposure.

Responsibilities:

- Analyze technical information (diagrams, Entity-Relationship schemas, tools) about the customer's business model to be explained to all team members.
- Understanding legacy process (Simple Object Access Protocol messages using ESB Tibco) and tech stack (IBM Master Data Management, IBM Operational Decision Manager) to propose new solutions at the microservices level using Java as tech stack.
- Use technical specifications to apply them to new in-house developments.
- Understand legacy Java code to migrate to the new microservices structure.
- Understand functional testing used by the QA area to help to create more testing scenarios using SoapUI.
- Identify development best practices and code standards applied in MoneyGram to be implemented by our team members.
  - Applied RSA for integration modeling and architecture documentation.
  - Collaborated on migration from Tibco ESB to Java microservices with IIB/ACE exposure.

Environment: Spring Boot, Java 8, Layer 7, Tibco (ESB), MDM (IBM), ODM (IBM), Maven, SoapUI, Oracle 12g, Couchbase (NoSQL), Bitbucket, Jenkins, Eclipse, SOAP (Big Services), Scrum and Kanban boards, Jira, Windows.

Social Development Ministry, Mexico City- Grupo Altavista **Apr 2016 – Sep 2018**

Role: Middleware Engineer

Description: The Social Development Ministry has web applications to accomplish its mission based on the National Development Plan. This infrastructure includes the first approach to distributed systems.

Responsibilities:

- Administrate services in JBoss Enterprise Application Platform (EAP) v5 & v6.
- Administrate Linux boxes installed on Red Hat Enterprise Linux (RHEL) v6 & v7.
- Responsible for developing, maintaining, performing, and monitoring the middleware environment using JBoss and RHEL, creating Linux boxes, configuring tools inside Linux boxes, creating users, and modifying permission access.
- Debug, resolve, and optimize code's developers to be deployed on JBoss reading code and using Visual VM.
- Identify, research, and resolve defects within a given application.
- Identify JVM thread dumps and memory leaks using IBM Thread and Monitor Dump Analyzer for Java (TDMA).

Gerardo

- Guide QA Team to implement test scenarios like availability, performance, code quality, response time, and usability using tools such as JMeter and SonarQube.
- Use bash scripting to create an automated deployment process. First contact with DevOps philosophy: Jenkins, Red Hat OpenShift, Red Hat Ansible, Kubernetes.
- Linux system administration.
- Manage Red Hat Enterprise Virtualization (RHEV) console.
- Use Apache Subversion (SVN) as the control versioning tool.
- Know customer's business.

Environment: RHEL 6 & 7, JBoss EAP 5 & 6, Java 6 & 7, Java Mail API, Unix bash scripting, JSF, Seam Framework, JPA, Hibernate, Maven, Oracle 12g, Jenkins, Artifactory, JMeter, SonarQube (Check style, PMD, Find Bugs), IBM Thread and Monitor Dump Analyzer for Java (TDMA), Visual VM, Eclipse, SVN, Fedora Distribution.

Supremo Consejo de México, Mexico City- Artifex Tektone **Mar 2016 – Mar 2016**
Role: Java Architect

Description: The executive board needs control over the participants' inscription to the next congress. Therefore, my objective was to create a complete portal to register the congressman and offer administrative information.

Responsibilities:
- Create the complete website.
- Generate the Entity-Relationship (12 tables) diagram.
- Database mapping (Hibernate) and implementation using public MySQL web hosting.
- Design web application (HTML, JSP) architecture using security and MVC patterns with Spring Security and Spring MVC.
- Developing and implementing the system using public web hosting.
- Generate executive information using the Java API for Microsoft Documents (Apache POI).
- Use GitHub as the control versioning tool.
- Use Maven as software project management.
- Technical support.

Environment: Java 8, Spring MVC 4.0, Spring Security, Hibernate, JSP, Apache POI, HTML, jQuery, AJAX, JSON, JavaScript, CSS, Maven, MySQL, MySQL Workbench, Tomcat, Eclipse, Git, GitHub, Mac OS X.

Lithomat, Mexico City-Artifex Tekton **May 2014 – Feb 2016**
Role: Java Architect

Description: The CEO of the print shop needs to automate its entire business process with Information Technology. My objective was to create an Enterprise Resource Planning (ERP) to allow the management of the print shop. Generate executive information on stocks, sales, clients' accounts, accounts receivable and accounts payable, human resources, and maintenance. Generate reports in Excel and PDF.

Responsibilities:

Gerardo

- Interview with stakeholders to get functional and non-functional requirements and describe business logic flow using Unified Modeling Language (UML) diagrams.
- Understanding patterns of the production process.
- Generate and implement an Entity Relationship diagram (64 tables) to accomplish the functional requirements using
- MySQL and MySQL Workbench.
- Design web application architecture using Java technology and MVC pattern and security (Spring MVC, Spring Security, Hibernate, JSP, HTML, raw JavaScript, CSS).
- Implement the system into a local Mac OS X server.
- Use GitHub as the code repository.
- Generate executive reports used by the CEO through a secured module using Jasper Reports.
- Technical support.

Environment: Java 6, Spring MVC 4.0, Spring Security, Hibernate, JSP, Servlets, Jasper Reports, HTML, jQuery, AJAX, JSON, JavaScript, CSS, Maven, MySQL, MySQL Workbench, Tomcat, Eclipse, Git, GitHub, Mac OS X.

Low-cost airline Interjet, Mexico City- Artifex Tekton **May 2012— Apr 2014**
Role: Sr Java Developer

Description: The marketing office needs to generate brand presence in social networks and obtain its clients' possible consumer patterns. Therefore, my objective was to create multiple web applications to get the airline's quality feedback and create numerous web applications to reward the client's loyalty.

Responsibilities:
- Design, develop, and implement multiple Entity relationship diagrams.
- Design, develop, and implement multiple web application architectures.
- Generate manual executive information from marketing quizzes using in-house queries.
- Technical support.
- Environment: Java 6, Spring MVC 4.0, Spring JDBC, JSF, HTML, AJAX, JavaScript, CSS, MySQL, Tomcat, JDeveloper, Mac OS X.

Department of the Treasury, Mexico City, Department of the Treasury **Oct 2009 – Apr 2012**
Role: Web Java Developer

Description: Federal Treasury daily admits hundreds of payment declarations sent to the Federal Tax Administration service. My objective was to analyze and improve code from the legacy Swing application. Develop a new web application using Spring MVC technology. Unify the business logic of separated modules related to creating and retrieving payment declarations into a unique web application. Generate further crossed information from payment declarations. Maintain levels of information security and encryption required by the law.

Responsibilities:
- Understanding Federal Treasury business logic.
- Debug the legacy Java Swing applications to get the high-level requirements.
- Understanding steps are done to encrypt data in-house.
- Develop improvements in database schema adding tables, indexes, and triggers.

Gerardo

- Developing web application using the J2EE framework: JSP and the first contact with Spring MVC.
- Maintenance of legacy applications until production lives.
- Technical leader.
- Technical support.

Environment: Java 6, Java Swing, Spring MVC 3.0, JSP, HTML, AJAX, JavaScript, CSS, MySQL, BEA WebLogic Application Server, Oracle, JDeveloper, Windows.

Argos Afore, Mexico City- Pagoss Consultory **Nov 2008 – Sep 2009**
Role: Java Developer

Description: Argos Afore is a financial institution responsible for pension savings management. My objective was to give continuity to the business by developing improvements related to the view: JSP, Servlets. Create new reports using XSLT in the XML data. Improve database queries.

Responsibilities:
- Learn mastering concepts such as Extensible Markup Language (XML) and Extensible Stylesheet Language Transformations (XSLT).
- Use the Java API from XML Processing (JAXP) to manage XML documents using Document Object Model (DOM), Simple API for XML (SAX), and XSLT.
- Learn Apache Struts (struts.xml) framework to implement the Model-View-Controller pattern.
- Use the Java Server Pages Tag Library (JSTL) to encapsulate functionality related to reporting financial data expressed in XML format.
- Understanding business concepts to understand the Entity-Relationship diagram and create complex queries to represent new reports.
- Working with Stored Procedures.
- Manual testing.

Environment: Java 5, Struts, JSP, JSTL, XML, XSLT, JAXP, DOM, SAX, HTML, JavaScript, CSS, Oracle, Windows.

Elektra, Mexico City- Elektra **May 2008 – Oct 2008**
Role: Java Developer

Description: Elektra is a Mexican retail store with many stores in the country. My objective was to help senior programmers make maintenance in-house web applications used by the Human Resources department.

Responsibilities:
- Learning mastering concepts as Hypertext Transfer Protocol (HTTP), Servlets (javax. servlet, and javax. servlet. Http packages), JavaServer Pages (JSP), Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), JavaScript.
- Learning Servlet Lifecycle: init(), service (), doGet (), doPost(), doHead(), doPut(), doDelete(), destroy ().
- First contact with servlets/JSP in a web application (web.xml) that does not contain the Model-View-Controller pattern.

Gerardo

- Use filters (web.xml + javax. servlet. Filter) to validate authentication and authorization of request for resources.
- Maintenance of the JSP/servlets used at the web project related to evaluating employees.
- Use Java Database Connection (JDBC) to get data from a database.
- Technical support.

Environment: Java 4, HTTP, Servlets, JSP, JDBC, Windows.

Medica Sur Hospital, Mexico City-Medica Sur Hospital **Jul 2006 – Apr 2008**
Role: Programmer

Description: The Hospital administration is very complex. There are many data subdomains related to each other to give a complete health service required by the patient. My objective was to help senior programmers extend the modules implemented in the in-house Electronic Health Record (EHR). This EHR was built in OpenEdge Advanced Business Language (Progress 4GL).

Responsibilities:

- Mastering the concepts of Relational Database.
- Create a complete module to help schedule patients at the Radiology Department. This module includes the development of a new Entity-Relationship diagram of 32 tables.
- Generate a new Graphical User Interface (GUI) to manage the Scheduling module.
- Improve queries to consume data already populated in other tables unrelated to the ER diagram.
- Technical support.

Environment: Progress 4GL, Windows.